

Code No: R2321051

II B. Tech I Semester Regular Examinations, November- 2024
DISCRETE MATHEMATICS & GRAPH THEORY

(Common to

CSE, IT, CSE(AI&ML), CSE(AI), CSE(DS), CSE(AI&DS), CSE(CS), CSE(IOT&CSIBCT),
 CSE(CS&BS), CS&BS, AI&DS, CS, AI&ML, CS&D, COS)

Max. Marks: 70

Time: 3 hours

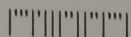
- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Form the conjunction of (2M)
 P : it is raining today.
 Q : There are 2 tables in this room.
- b) Prove $(P \rightarrow Q) \Leftrightarrow (\neg P \vee Q)$. (2M)
- c) Show that $A \subseteq B \Leftrightarrow A \cup B = B$. (2M)
- d) Let $N = \{1, 2, 3, \dots\}$ be ordered by divisibility. State whether each of the following subsets of N are linearly (totally) ordered. (2M)
 (i) $\{24, 2, 6\}$; (ii) $N = \{1, 2, 3, \dots\}$; (iii) $\{7\}$
- e) In how many ways can we draw a heart or a spade from an ordinary deck of playing cards (2M)
- f) In how many ways can 7 women and 3 men be arranged in a row if the 3 men must always stand next to each other? (2M)
- g) Is there a graph with degree sequence $(1, 3, 3, 3, 5, 6, 6)$?, why? (2M)
- h) Give the simplest possible example of a non null, non directed graph with (2M)
 (i). having no vertices of odd degree.
 (ii). having no vertices of even degree.
- i) Define Bipartite and Planar Graph. (2M)
- j) Define Euler graph. (2M)

PART -B (5x10Marks=50M)**UNIT-I**

2. a) Obtain disjunctive normal forms of (10M)
 (i) $P \wedge (P \rightarrow Q)$ (ii) $\neg(P \vee Q) \Leftrightarrow (P \wedge Q)$.
 OR
- b) Show that $R \vee S$ follows logically from the premises $C \vee D, (C \vee D) \rightarrow \neg H, \neg H \rightarrow (A \wedge \neg B)$, and $(A \wedge \neg B) \rightarrow (R \vee S)$. (10M)

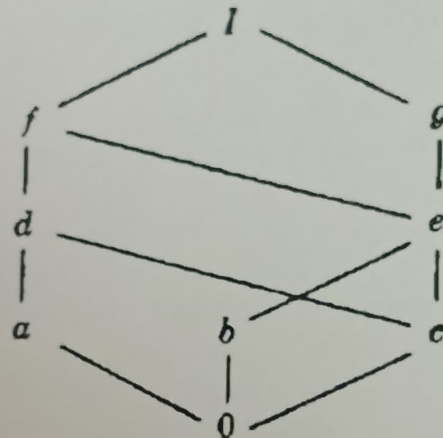


UNIT-II

3. a) Let A be a given finite set and $\rho(A)$ its power set. Let $\subseteq$ be the inclusion relation on the elements of $\rho(A)$. Draw Hasse diagrams of $(\rho(A), \subseteq)$ for (i) $A = \{a\}$ (ii) $A = \{a, b\}$ (iii) $A = \{a, b, c\}$ (iv) $A = \{a, b, c, d\}$. (10M)

OR

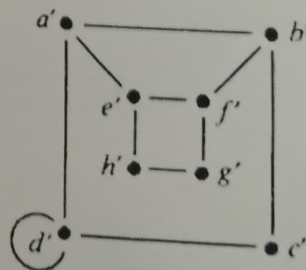
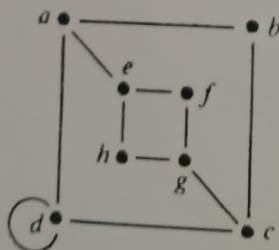
- b) Consider the lattice M in the following figure (10M)
- Find all join-irreducible elements.
 - Find the atoms.
 - Find complements of a and b , if they exist.
 - Express each x in M as the join of irredundant join-irreducible elements.
 - Is M distributive? Complemented?

UNIT-III

4. a) Determine the coefficient of $x_1^3 x_2^2 x_3^2 x_5^5$ in $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$. (10M)
- OR
- b) Solve $a_n - 6a_{n-1} + 8a_{n-2} = 3^n$ where $a_0 = 3$, and $a_1 = 7$. (10M)

UNIT-IV

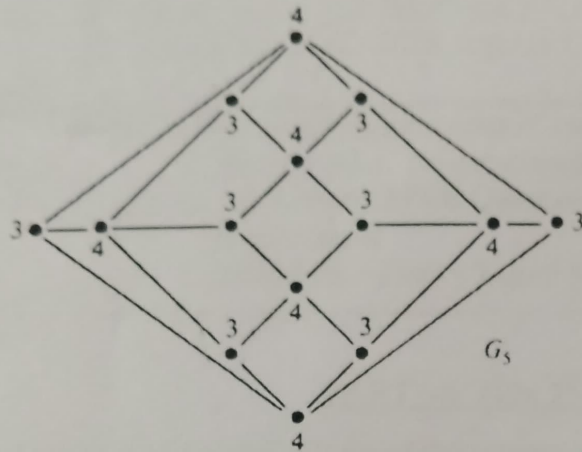
5. a) Show that the two graphs in figures are not isomorphic. (10M)



OR

- b) Show that there is no Hamiltonian cycle for the following graph G_5 .

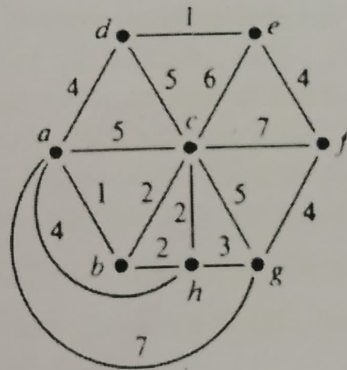
(10M)

UNIT-V

6. a) Prove that a connected graph G is a tree if and only if G has fewer edges than vertices. (10M)

OR

- b) Find a minimal spanning tree for the following graph. (10M)



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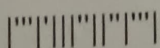
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PART -A (10x 2Marks=20M)

1. a) Translate into symbolic form of the statement: (2M)
Jack and Jill went up the hill.
- b) Write the duals of $(P \vee Q) \wedge R$. (2M)
- c) If $S = \{a, b, c\}$, find nonempty disjoint sets A_1 and A_2 such that $A_1 \cup A_2 = S$. (2M)
Find other solutions to this problem.
- d) Let $N = \{1, 2, 3, \dots\}$ be ordered by divisibility. State whether each of the (2M)
following subsets of N are linearly (totally) ordered.
(i) $\{3, 15, 5\}$; (ii) $\{2, 8, 32, 4\}$ (iii) $\{15, 5, 30\}$
- e) How many ways can we get a sum of 4 or of 8 when two distinguishable dice (2M)
(say one die is red and the other is white) are rolled?
- f) In how many ways can the letters of the English Alphabet be arranged so that (2M)
there are exactly 5 letters between the letters a and b ?
- g) Is there a non-simple graph G with degree sequence $(1, 1, 3, 3, 3, 4, 6, 7)$?, why? (2M)
- h) Give the simplest possible example of a non null, non directed graph with (2M)
(i). having no vertices of even degree.
(ii). having exactly one vertex of odd degree;
- i) State Euler's theorem. (2M)
- j) What is meant by a minimal spanning tree? (2M)

PART -B (5x10Marks=50M)UNIT-I

2. a) Obtain a conjunctive normal form of (10M)
(i) $P \wedge (P \rightarrow Q)$ (ii) $\neg(P \vee Q) \Leftrightarrow (P \wedge Q)$
OR
- b) Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$. (10M)

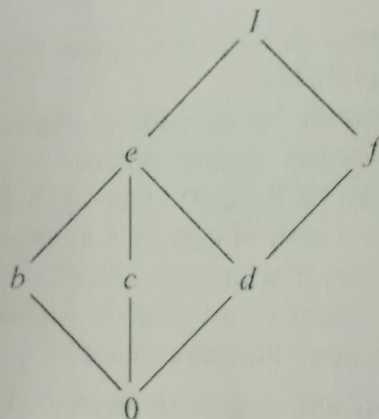


UNIT-II

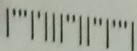
3. a) Give an example of a nonempty set and a relation on the set that satisfies each of the following combinations of properties, draw a digraph of the relation. (10M)
- Symmetric and transitive, but not reflexive.
 - Symmetric and reflexive, but not transitive.
 - Transitive and reflexive, but not symmetric.
 - Transitive and reflexive, but not antisymmetric.
 - Transitive and antisymmetric, but not reflexive.

OR

- b) Consider the bounded lattice L in shown in the following figure. Then (10M)
- Find the complements, if they exist, of e and f .
 - Express I in an irredundant join-irreducible decomposition in as many ways as possible.
 - Is L distributive?
 - Describe the isomorphisms of L with itself.

UNIT-III

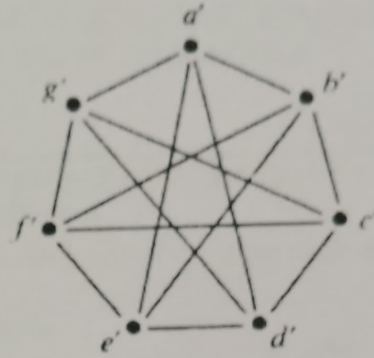
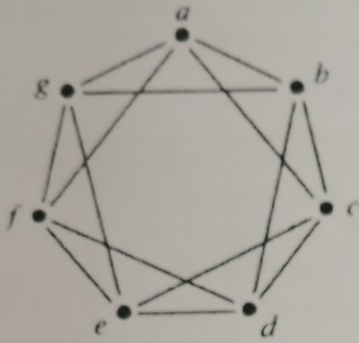
4. a) Determine the coefficient of $x^5 y^{10} z^5 w^5$ in $(x - 7y + 3z - w)^{25}$. (10M)
- OR
- b) Find the form of a particular solution to $a_n - 5a_{n-1} + 6a_{n-2} = n^2 4^n$ for $n \geq 2$. (10M)



UNIT-IV

5. a) Show that the following graphs are isomorphic.

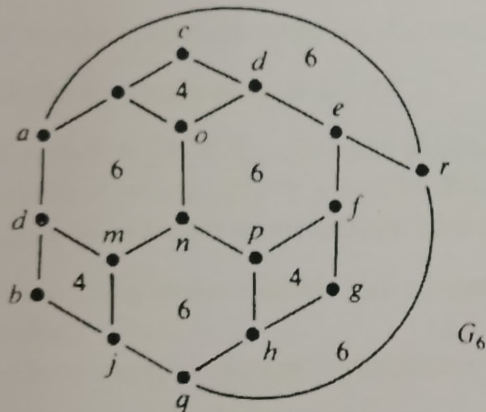
(10M)



OR

- b) Show that the following graph G_6 has no Hamiltonian cycle.

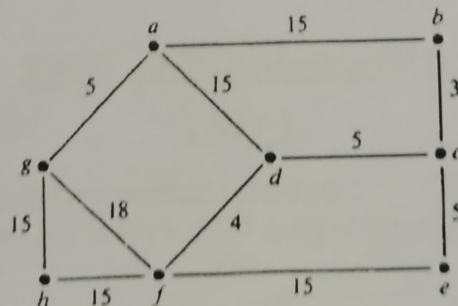
(10M)

UNIT-V

6. a) Suppose that G is a connected graph. Prove that any circuit and the complement of any spanning tree have an edge in common. Is the complement of a spanning tree also a spanning tree? (10M)

OR

- b) Determine a railway network of minimal cost for the cities in the following Figure. (10M)



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PART -A (10x 2Marks=20M)

1. a) Construct the truth table for the statement formula $P \vee \neg Q$. (2M)
- b) Write the duals of $(P \wedge Q) \vee T$. (2M)
- c) Given $S = \{1, 2, 3, \dots, 10\}$ and a relation R on S , (2M)
 where $R = \{\langle x, y \rangle | x + y = 10\}$. What are the properties of the relation R ? (2M)
- d) Give an example of an infinite lattice L with finite length. (2M)
- e) How many ways can we get a sum of 8 when two Indistinguishable dice are rolled? (2M)
- f) In how many ways can 5 children arrange themselves in a ring? (2M)
- g) Draw a picture of the following graph and state whether it is directed or non-directed and whether it is simple. $G_1 = (V_1, E_1)$, where (2M)
 $V_1 = \{\{a, b\}, \{b, c\}, \{a, c\}, \{a, d\}, \{d, e\}\}$
- h) Give the simplest possible example of a non null, non directed graph with (2M)
 (i). having exactly one vertex of even degree.
 (ii).having exactly two vertices of odd degree; (2M)
- i) What is a planar graph. (2M)
- j) Draw the graph of $K_{2,5}$. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Obtain the principal disjunctive normal forms of (10M)
 (i) $\neg P \vee Q$ (ii) $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$.
 OR
- b) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \wedge Q, Q \rightarrow R, P \rightarrow M$, and $\neg M$. (10M)

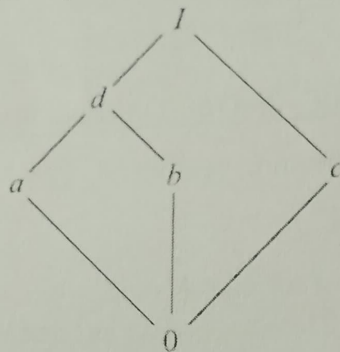
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UNIT-II

3. a) Let $X = \{a, b, c, d, e\}$ and let $C = \{\{a, b\}, \{c\}, \{d, e\}\}$. Show that the partition C defined an equivalent relation on X . (10M)

OR

- b) Consider the bounded lattice L in the following figure then (10M)
- Find the complements, if they exist, of a and c .
 - Express I in an irredundant join-irreducible decomposition in as many ways as possible.
 - Is L distributive?
 - Describe the isomorphisms of L with itself.

UNIT-III

4. a) Find the coefficient of $x_1^2 x_3 x_4^3 x_5^4$ in the expansion of $(x_1 + x_2 + x_3 + x_4 + x_5)^{10}$. (10M)

OR

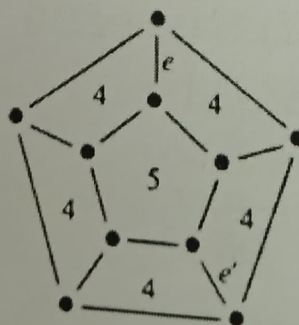
- b) Find a particular solution to $a_n - 7a_{n-1} + 10a_{n-2} = 7 \cdot 3^n + 4^n$. (10M)

UNIT-IV

5. a) Prove that a graph G is isomorphic to a graph H if there is an ordering of the vertices of G and H such that the resulting adjacency matrices are equal. (10M)

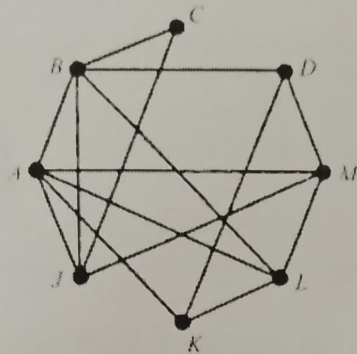
OR

- b) The graph G_7 (shown in the following figure) does possess Hamiltonian cycles. Show, however, that any such cycle containing one of the edges e, e' must avoid the other. (10M)

 G_7

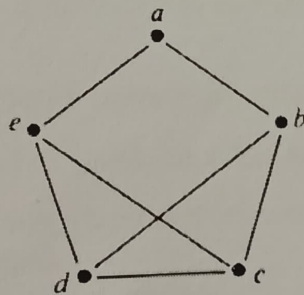
UNIT-V

6. a) Define what is meant by chromatic number and find the chromatic number n of the following graph. (10M)



OR

- b) Determine the different non isomorphic spanning trees for the following graph. (10M)



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from **Part-B**4. All Questions carry **Equal Marks****PART -A (10x 2Marks=20M)**

1. a) Construct the truth table for $(P \vee Q) \vee \neg P$. (2M)
- b) Write the duals of $\neg(P \vee Q) \wedge (P \vee \neg(Q \wedge \neg S))$. (2M)
- c) Give an example of a set X such that $\langle \rho(X), \subseteq \rangle$ is a totally ordered set. (2M)
- d) Let D_m denote the positive divisors of m ordered by divisibility. Draw the Hasse diagram of D_{12} . (2M)
- e) How many 3-digit numbers can be formed using the digits 1,3,4,5,6,8,and 9? (2M)
- f) In how many ways can a hand of 5cards be selected from a deck of 52 cards? (2M)
- g) Draw a picture of the following graph and state whether it is directed or non-directed and whether it is simple.
 $G_2 = (V_2, E_2)$, where $V_2 = \{a, b, c, d, e\}$ and
 $E_2 = \{(a, b), (a, c), (a, d), (a, e), (e, c), (c, a)\}$. (2M)
- h) What is the largest possible number of vertices in a graph with 35edges and all vertices of degree at least3? (2M)
- i) Define Bi-partite graph. (2M)
- j) Define chromatic number. (2M)

PART -B (5x10Marks=50M)**UNIT-I**

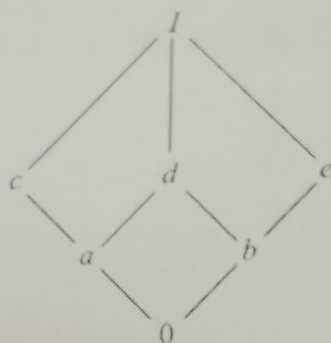
2. a) Obtain the principal conjunctive normal form of the formula S given by (10M)
 $(\neg P \rightarrow R) \wedge (Q \rightleftharpoons P)$.
OR
b) Show that $R \rightarrow S$ can be derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$, and Q . (10M)

UNIT-II

3. a) Prove that if R is a transitive and irreflexive relation on a set A , then R is antisymmetric and asymmetric. (10M)

OR

- b) Consider the bounded lattice L in the following figure then (10M)
- Find the complements, if they exist, of a and c .
 - Express I in an irredundant join-irreducible decomposition in as many ways as possible.
 - Is L distributive?
 - Describe the isomorphisms of L with itself.

UNIT-III

4. a) Use the binomial theorem to prove that $3^n = \sum_{r=0}^n \binom{n}{r} 2^r$. (10M)

OR

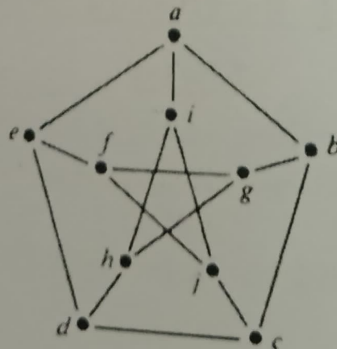
- b) Find a particular solution to $a_n - 7a_{n-1} + 10a_{n-2} = 7 \cdot 3^n$ for $n \geq 2$. (10M)

UNIT-IV

5. a) Prove that if G is a simple nondirected graph with 6 vertices, then G or the complement of G contains 3 mutually adjacent vertices. (10M)

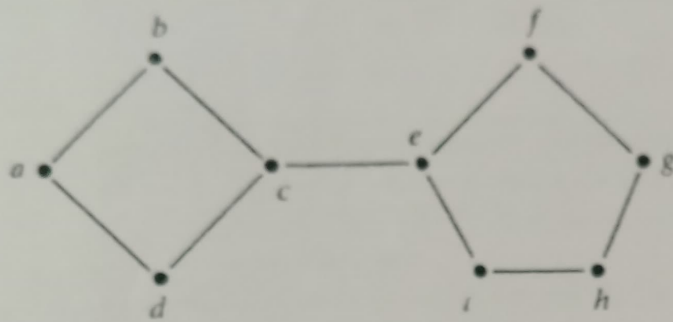
OR

- b) Prove that there is no Hamiltonian cycle for the following graphs: (10M)



UNIT-V

6. a) Describe how to obtain all possible spanning trees for the following graphs and list the total number of different spanning trees: (10M)



OR

- b) Prove that a wheel graph W_n , where n is odd, is a 4-critical graph (10M)

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R23

SET - 1

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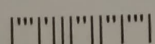
PART -A (10x 2Marks=20M)

1. a) Define Human Values. (2M)
- b) Distinguish between Happiness and Excitement. (2M)
- c) List out the activities of self. (2M)
- d) What do you understand by trust? (2M)
- e) What is the outcome of harmony with in the whole of nature/existence? (2M)
- f) List any three forms of disrespect. (2M)
- g) Write the four components to ensure humanistic tradition. (2M)
- h) What is preconditioning? Explain. (2M)
- i) Assess how Utility value and Artistic value are related. (2M)
- j) Describe the guidelines of professional ethics. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Describe the process of self-exploration with the help of a diagram. Give one example from your life to illustrate how you self-explored any proposal shared in the class. (10M)
- OR
- b) Describe Happiness. Explain how mutual fulfillment is essential in human relationship. (10M)



UNIT-II

3. a) Distinguish between the response of the Self and Body by giving any two examples. (10M)

OR

- b) Describe the activities of imagination in the Self. Justify how imagination gets enslaved with lack of right understanding? Explain with a few examples. (10M)

UNIT-III

4. a) Explain respect. How are we generally making differentiation in the name of respect? Explain the minimum content of respect for a human being (10M)

OR

- b) Describe the four human goals in the society and how are they related? With this at the background, evaluate the current state of society today. (10M)

UNIT-IV

5. a) Process in nature are cyclical and mutually enriching. Explain this statement with 3 examples. (10M)

OR

- b) Explain the submergence of nature in to space. How is this related to a student going through a professional course? Analyze keeping yourself at the focus and explain. (10M)

UNIT-V

6. a) Explain the term 'holistic technology'. Take any two such examples from the technologies developed in Indian history and evaluate if they are holistic. (10M)

OR

- b) Analyze how human conduct, education, constitution and order are related. How do they ensure the human tradition? Explain with reference to the present state of society. (10M)

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SET - 2

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PART -A (10x 2Marks=20M)

1. a) State out the two domains of education. (2M)
- b) What do you mean by your natural acceptance and experiential validation? (2M)
- c) List out four orders in nature. (2M)
- d) What is the outcome of harmony with the whole of nature/existence? (2M)
- e) What do you mean by continuity of Happiness? (2M)
- f) Define the terms Gathansheel, Gathanpurana, Kriyapurnata and Acharanpurnata. (2M)
- g) Write a short note on Nature (Units) submerged in space. (2M)
- h) What are basic human aspirations? Explain. (2M)
- i) Define the terms Swatantra and Swarajya. (2M)
- j) What do you mean by sukh and suvidha? (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Explain the basic guidelines for Value Education. Analyze why are they important? How does the process of value education emerge from the guidelines? (10M)
- OR
- b) Discuss the basic guidelines for value education? Differentiate between prosperity and wealth with examples. (10M)

Code No: R2321012

R23

SET -2

UNIT-II

3. a) Distinguish between the needs of the self (I) and the needs of the body.

OR

(10M)

- b) Explain how dosensations and pre conditioning's influence our imagination?
Give two examples of each.

(10M)

UNIT-III

4. a) Define the concept of justice. How does it lead to mutual happiness?

OR

(10M)

- b) Explain respect. How do we are generally differentiate in the name of respect? Explain the minimum content of respect for a human being.

(10M)

UNIT-IV

5. a) Discuss in detail the four orders in nature? How are they inter-dependent and mutually fulfilling for each other? Explain.

OR

(10M)

- b) What do you mean by innateness? Describe the innateness in the four orders?

(10M)

UNIT-V

6. a) What are the values in interaction of human beings with the material things?
Give one example of each.

OR

(10M)

- b) State the concept of professional ethics. Explain it in detail.

(10M)

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Max. Marks: 70

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2. Answer **ALL** the question in **Part-A**
3. Answer any **FIVE** Questions, each Question from each unit
from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) The need for physical facilities is temporary. Explain. (2M)
- b) What should be the aim of competence in professional ethics? (2M)
- c) Write the difference between sanyam and care. (2M)
- d) State the need and importance of ethics in the life of an individual. (2M)
- e) What is the main reason that one should go for self exploration? (2M)
- f) Compare the inheritance of Bio order and Physical order. (2M)
- g) Define the word value and illustrate with one example. (2M)
- h) Briefly describe the natural characteristics of the Animal Order. (2M)
- i) List the two forms of disrespect. (2M)
- j) Distinguish between Intention and Desire. (2M)

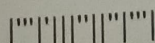
PART -B (5x10Marks=50M)

UNIT-I

2. a) Discuss your present vision of a happy and prosperous life. (10M)
- OR
- b) Describe the difference between belief and understanding with a neat table. (10M)

UNIT-II

3. a) Why are physical facilities required? What do you mean by right utilization of the body? (10M)
- OR
- b) Write a detailed note on understanding the harmony in self. (10M)



Code No: R2321012

R23

SET - 3

UNIT-III

4. a) "Family is a natural laboratory to understand human relationships"-elaborate. (10M)

OR

- b) Discuss the basis of an undivided society-the world family. Explain in detail. (10M)

UNIT-IV

5. a) Closely examine four orders in nature and their participation. Explain present state of participation of human order. (10M)

OR

- b) Write about recyclability and self-regulation in nature with examples. (10M)

UNIT-V

6. a) How do human values affect the life of professionals? Illustrate. (10M)

OR

- b) Define the term ethical human conduct. Explain it in terms of values, policies, and character. (10M)

Code No: R2321012

R23

SET - 4

II B. Tech I Semester Regular Examinations, November- 2024
UNIVERSAL HUMAN VALUES-UNDERSTANDING HARMONY AND ETHICAL
HUMAN CONDUCT

(Common to
CE,EEE,ME,ECE,CSE,IT,AME,MM,AGE,CSE(AI&ML),CSE(AI),CSE(DS),CSE(AI&DS),
CSE(CS),CSE(IOT&CSIB,CT),CSE(CS&BS),FE,AI&DS,Ph.E,CS&BS,CS,AI&ML,CS&D,
COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
3. Answer any **FIVE** Questions, each Question from each unit
from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Mention the four dimensions of a human being. (2M)
- b) How do you presently decide what is valuable to you? How do you ensure that your decision is right? (2M)
- c) Write down two consequences when we assume "Human being = Body". (2M)
- d) If someone is helping to fulfil your needs in terms of body and self, write feelings for them. (2M)
- e) Write the two sentences about respect. (2M)
- f) Write any two sentences about yourself exploration in present days. (2M)
- g) Discuss the concept of Experimental validation. (2M)
- h) Define the term holistic technology. (2M)
- i) Define the terms utility value and artistic value. (2M)
- j) List down the four components to ensure Humanistic Tradition. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) How you look at your basic aspirations. Is it 'To Be', 'To Get', 'To Become'? Justify your right priority with examples. (10M)
OR
- b) Explain Need, Basic guidelines of value education. (10M)

UNIT-II

3. a) Explain body as an instrument of 'I'. (10M)
OR
- b) Harmony in 'I' means understanding characteristics and activities of 'I'. Explain. (10M)

|||||

UNIT-III

4. a) For a fearlessness in society, what are minimum requirements? Where do you wish to live, fearful or fearless society? What is your participation towards ensuring fearless society, examine within yourself? (10M)

OR

- b) Comment, Human to Human relationship 'Is' or being 'Created'. Explain feeling of 'Love' in light of above statement. (10M)

UNIT-IV

5. a) Explain the feeling of Care and Guidance. What role do these feelings have in parents child relationship? (10M)

OR

- b) What do you mean by innateness? What is the innateness in the four orders? (10M)

UNIT-V

6. a) What do you understand by Profession? How do you apply the understanding you got in this course to ensure ethical living? (10M)

OR

- b) Explain the term 'holistic technology'. Take any two such examples from the technologies developed in Indian history and evaluate if they are holistic. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
ADVANCED DATA STRUCTURES AND ALGORITHMS ANALYSIS

(Common to
 CSE, CSE(AI&ML), CSE(AI), CSE(DS), CSE(AI&DS), CSE(CS), CSE(IOT&CSIBCT),
 CSE(CS&BS), CS&BS, AI&DS, CS, AI&ML, CS&D, COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**
- ~~~~~

PART -A (10x 2Marks=20M)

1. a) Define big Oh notation. (2M)
- b) Define AVL tree. (2M)
- c) Define min-heap. (2M)
- d) What is convex hull? (2M)
- e) Define dynamic programming. (2M)
- f) Discuss briefly about Travelling Sales man Problem. (2M)
- g) Define LIFO branch and bound. (2M)
- h) State and explain graph coloring problem. (2M)
- i) Explain the concept of polynomial-time reduction between decision problems. (2M)
- j) Define NP hard problem. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) The running time of an algorithm is represented by the following recurrence relation (10M)

$$T(n) = \begin{cases} n & \text{if } n \leq 3 \\ T\left(\frac{n}{3}\right) + cn & \text{otherwise} \end{cases}$$

Find the time complexity of an algorithm
 OR

- b) Define B tree. Show how to insert and delete an element from B tree. (10M)

UNIT-II

3. a) Define Max-heap. Write Max-Heapify algorithm that maintain max-heap property. (10M)

OR

- b) Divide and conquer paradigm involves three steps at each level of recursion. What are all they? Show that merge-sort algorithm closely follows these steps. Illustrate the operation of merge sort on the array $A = \{3, 41, 52, 26, 38, 57, 9, 49\}$ (10M)

UNIT-III

4. a) What is Optimal Binary Search Tree? Explain in detail the implementation of OBST with an example. (10M)

OR

- b) State the Job – Sequencing with deadlines problem. Find an optimal sequence to the $n=5$ Jobs where profits $(P_1, P_2, P_3, P_4, P_5) = (20, 15, 10, 5, 1)$ and deadlines $(d_1, d_2, d_3, d_4, d_5) = (2, 2, 1, 3, 3)$. (10M)

UNIT-IV

5. a) Give the 0/1 Knapsack Branch and Bound algorithm. Explain how to find optimal solution using variable – tuple sized approach. (10M)

OR

- b) Solve the 8 queen problem using backtracking technique? Draw the state space tree? (10M)

UNIT-V

6. a) What is NP-complete problem and also explain how to solve NPC problem using reduction method. (10M)

OR

- b) State and explain Cooks theorem in detail. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
ADVANCED DATA STRUCTURES AND ALGORITHMS ANALYSIS

(Common to
 CSE, CSE(AI&ML), CSE(AI), CSE(DS), CSE(AI&DS), CSE(CS), CSE(IOT&CSIBCT),
 CSE(CS&BS), CS&BS, AI&DS, CS, AI&ML, CS&D, COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**
- ~~~~~

PART -A (10x 2Marks=20M)

1. a) Define B tree. (2M)
- b) Define Theta notation. (2M)
- c) What are biconnected components? (2M)
- d) What are the ways to represent a graph? (2M)
- e) Define greedy method. (2M)
- f) Give the example of string editing? (2M)
- g) Define backtracking. (2M)
- h) Define the FIFO branch and bound. (2M)
- i) Show that certain scheduling problems, like the job shop scheduling problem, are NP-hard. (2M)
- j) Define clique decision problem. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Show various stages of AVL tree on inserting the items MAR, MAY, NOV, AUG, APR, JAN, DEC, JUL and FEB. (10M)
- OR
- b) Solve the following recurrence: $T(n)=4T(n/2)+n$, Where $n \geq 1$ and is a power of 2. (10M)

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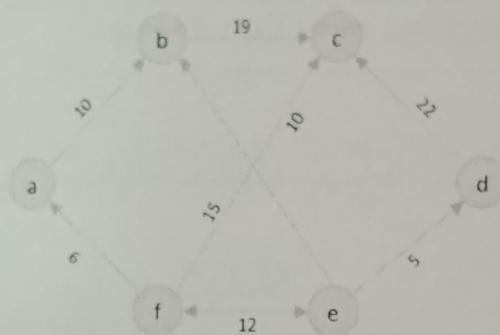
UNIT-II

3. a) Rearrange following numbers using quick sort: 10, 6, 3, 7, 17, 26, 56, 32, 72 (10M)
- OR

- b) Write and explain in brief Strassen's matrix multiplication with a suitable example (10M)

UNIT-III

4. a) Write an algorithm for Travelling Sales person problem and implement for the following graph. (10M)



OR

- b) State the Job – Sequencing with deadlines problem. Find an optimal sequence to the $n=5$ Jobs where profits $(P_1, P_2, P_3, P_4, P_5) = (10, 15, 20, 4, 2)$ and deadlines $(d_1, d_2, d_3, d_4, d_5) = (1, 1, 3, 2, 2)$. (10M)

UNIT-IV

5. a) What is backtracking? Find a solution to the 4-Queens problem using backtracking strategy. Draw the solution space using necessary bounding function. (10M)

OR

- b) Give the general method of Branch and Bound Algorithm. Explain it with an example. (10M)

UNIT-V

6. a) Define P, NP, NP complete and NP-Hard problems. Give examples of each. (10M)

OR

- b) Explain how Cook's theorem establishes the NP-completeness of SAT. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
ADVANCED DATA STRUCTURES AND ALGORITHMS ANALYSIS

(Common to
 CSE, CSE(AI&ML), CSE(AI), CSE(DS), CSE(AI&DS), CSE(CS), CSE(IOT&CSIBCT),
 CSE(CS&BS), CS&BS, AI&DS, CS, AI&ML, CS&D, COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Define Omega notation. (2M)
- b) What is the purpose of AVL tree? (2M)
- c) What are connected components? (2M)
- d) Define divide and conquer technique. (2M)
- e) Define minimal spanning tree. (2M)
- f) State all pair shortest path problem. (2M)
- g) Define 0/1 knapsack problem. (2M)
- h) Differentiate between LIFO and FIFO branch and bound. (2M)
- i) Define chromatic number decision problem. (2M)
- j) What is job shop scheduling? (2M)

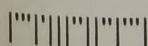
PART -B (5x10Marks=50M)

UNIT-I

2. a) Why do we use asymptotic notations in the study of algorithms? Briefly describe the commonly used asymptotic notations. (10M)
 OR
- b) Define B-Tree. Explain B-Tree of order m with an example. (10M)

UNIT-II

3. a) Write a Quick sort algorithm. Trace the algorithm and sort the elements 27, 87, 26, 61, 59, 17, 77, 53 (Show each step separately). Derive the Best-case time complexity of quick sort? (10M)
 OR
- b) Explain in detail about the binary search traversal with an example. (10M)



Code No: R2321053

UNIT-III

4. a) Construct an optimal binary search tree for the following items with probabilities given in the table below. (10M)

Items	1	2	3	4	5
Probability	0.24	0.22	0.23	0.3	0.01

OR

- b) Consider 5 items along with their respective weights and values $I = \langle I_1, I_2, I_3, I_4, I_5 \rangle$; $w = \langle 5, 10, 20, 30, 40 \rangle$; $v = \langle 30, 20, 100, 90, 160 \rangle$; The capacity of knapsack $W = 60$. Find the solution to the fractional knapsack problem (10M)

UNIT-IV

5. a) Give the statement of sum-of subsets problem. Find all sum of subsets for $n=4$, $(w_1, w_2, w_3, w_4) = (11, 13, 24, 7)$ and $M=31$. Draw the portion of the state space tree using fixed-tuple sized approach (10M)

OR

- b) Given a set of 4 items, each with weight $\{2, 2, 4, 5\}$ and benefit $\{3, 7, 2, 9\}$, Using LC branch and bound determine the items to include in the knapsack so that the total weight is less than or equal to a given weight limit and the total benefit is maximized. The weight limit for this knapsack is 10? (10M)

UNIT-V

6. a) Define the set of NP complete languages. Prove that travelling salesman problem is NP complete. (10M)

OR

- b) Prove, If any NP-complete problem belongs to class P, then is $P = NP$? (10M)

II B. Tech I Semester Regular Examinations, November- 2024
ADVANCED DATA STRUCTURES AND ALGORITHMS ANALYSIS
 (Common to

CSE, CSE(AI&ML), CSE(AI), CSE(DS), CSE(AI&DS), CSE(CS), CSE(IOT&CSIBCT),
 CSE(CS&BS), CS&BS, AI&DS, CS, AI&ML, CS&D, COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Define little omega notation. (2M)
- b) What are the properties of B tree? (2M)
- c) Compare connected components with bi-connected components. (2M)
- d) Define bellman ford algorithm. (2M)
- e) What is job sequencing with deadlines? (2M)
- f) List the steps involved in Dynamic programming. (2M)
- g) Discuss the statement of traveling salesman problem. (2M)
- h) What are the additional features required in branch-and-bound when compared to backtracking? (2M)
- i) Define NP Complete problems. (2M)
- j) State and explain cooks theorem. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Express the function $f(n) = 5n^3 + 5n^2 + 10n$ in Θ notation (10M)
 OR (10M)
- b) Build the AVL tree for the following list
 25, 30, 35, 40, 50, 55, 60, 65, 70, 75

UNIT-II

3. a) Define a Max Heap. Construct a max heap for the following
 12, 15, 9, 8, 10, 18, 7, 20, 25. (10M)
 OR

Code No: R2321053

- b) Sort the following list of elements by using Quick sort
99, 3, 7, 10, 2, 89, 1, 47, 5, 4.

(10M)

UNIT-III

4. a) Illustrate the concept of greedy algorithm and show how it is applied in solving the single source shortest path problem.

(10M)

OR

- b) With the help of dynamic programming algorithm explain all pair shortest path problem.

(10M)

UNIT-IV

5. a) Give an example to explain the travelling sales man problem using LC-branch and bound.

(10M)

OR

- b) Solve the graph coloring problem using backtracking technique. Draw the state space tree.

(10M)

UNIT-V

6. a) Discuss techniques for solving NP-hard graph problems, such as approximation algorithms or heuristics.

(10M)

OR

- b) Discuss approaches for tackling NP-hard scheduling problems, such as integer linear programming formulations or constraint programming.

(10M)

DIGITAL LOGIC & COMPUTER ORGANIZATION

(Common to
Time: 3 hours

(Common to CSE,IT, CSE(CS) CSE(IOT&CSIBCT),CSE(CS&BS),CS&BS,CS, CS&D,COS)
Time: 3 hours

Note: 1. Question Paper consists of two parts. (Part-I & Part-II)
2. Answer ALL questions.

Max. Marks: 70

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Explain the process of converting a decimal number (e.g., 97) to binary and vice versa. (2M)
- b) Why are NAND and NOR gates called universal gates? (2M)
- c) Distinguish between sequential circuits and combinational circuits. (2M)
- d) Discuss the operation of an SR flip-flop with a circuit diagram and truth table. (2M)
- e) Explain the process of adding two signed binary numbers using 2's complement representation. (2M)
- f) Define instruction-level parallelism. (2M)
- g) What is Read-Only Memory (ROM)? (2M)
- h) What is segmentation in memory management? (2M)
- i) What is the primary function of an interface circuit in I/O systems? (2M)
- j) What is the key difference between synchronous and asynchronous buses? (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Perform the addition of the following two signed binary numbers using 2's complement method: (10M)
- i) 1101_2 and 0110_2 ii) 1110_2 and 0111_2
- OR

- b) Simplify the following Boolean expression using Boolean algebra: (10M)
- i) $A'B + ABC' + ABC$ ii) $(BC' + A'D)(AB' + CD')$
- UNIT-II**

UNIT-II

3. a) What is a shift register? Design a 4-bit shift register and explain the difference between a left shift and a right shift operation. (10M)
- OR
- b) Explain the basic operational concepts of a computer with the fetch-execute cycle. (10M)
How do instructions get processed inside a computer?

Code No: R2321052

UNIT-III

4. a) What is a carry-look ahead adder? Explain its design and how it improves the speed of binary addition compared to a ripple-carry adder. (10M)

OR

- b) What is a hardwired control unit? Explain its design and operation. Compare it with a micro programmed control unit in terms of flexibility and speed. (10M)

UNIT-IV

5. a) Define cache hit and cache miss. How do cache hit ratios affect the overall performance of the memory system? Provide formulas used to measure cache performance. (10M)

OR

- b) Discuss the page replacement algorithms used in virtual memory systems. Which one is the most efficient in practice, and why? (10M)

UNIT-V

6. a) What is programmed I/O? Explain its operation and discuss the limitations of using programmed I/O in a computer system. (10M)

OR

- b) Compare the different data transfer methods: programmed I/O, interrupt-driven I/O, and DMA. Discuss the scenarios where each method is preferable. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
DIGITAL LOGIC & COMPUTER ORGANIZATION
 (Computer Science Engineering)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (Part-A and Part-B)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Explain the basic logic functions (AND, OR, NOT). Design a truth table for a 2-input AND gate. (2M)
- b) What is Excess-3 code? Convert the decimal number 249 to Excess-3 code. (2M)
- c) Give the functional units of a computer (2M)
- d) What factors affect the performance of a computer? (2M)
- e) Explain the process of adding two signed binary numbers using 2's complement representation with example (2M)
- f) Define instruction-level parallelism. (2M)
- g) Differentiate between internal and external fragmentation. (2M)
- h) What is segmentation in memory management? (2M)
- i) What is the difference between memory-mapped I/O and isolated I/O? (2M)
- j) Describe the role of bus in connecting I/O devices. (2M)

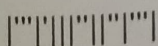
PART -B (5x10Marks=50M)

UNIT-I

2. a) Define fixed-point representation. Convert the decimal number 25.375 into fixed-point binary representation (both integer and fractional parts). (10M)
 OR
- b) Minimize the following Boolean function using a 3-variable K-map: (10M)
 $F(A,B,C) = \Sigma(0,1,2,5,7)$ and $F(A,B,C,D) = (0,1,2,4,5,7,11,15)$

UNIT-II

3. a) Explain how a ripple counter works. Design a 3-bit ripple down-counter using T flip-flops and explain its functioning. (10M)
 OR
- b) What is the difference between a multiprocessor and a multi-computer system? (10M)
 Discuss the advantages and challenges of using multiprocessor systems.
 Provide examples of their real-world applications.



UNIT-III

4. a) Explain the Booth's algorithm for multiplying signed binary numbers. (10M)

OR

- b) What is a multiple-bus organization in a computer system? Discuss the advantages of multiple-bus architecture compared to a single-bus system. (10M)

UNIT-IV

5. a) Explain the hierarchy of memory in computer systems. Discuss how different types of memory are organized and accessed in a typical computer system. (10M)

OR

- b) Explain the concept of virtual memory. How does virtual memory enable the execution of programs larger than the physical memory? (10M)

UNIT-V

6. a) What is the role of interrupts in I/O operations? Explain how interrupts work in a computer system and their significance in improving I/O efficiency. (10M)

OR

- b) What is Direct Memory Access (DMA)? Explain how DMA improves system performance by eliminating the need for CPU intervention during data transfers. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
DIGITAL LOGIC & COMPUTER ORGANIZATION
 (Computer Science Engineering)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Give examples of sequential circuits and combinational circuits. (2M)
- b) Convert the hexadecimal number 2F4A to binary, octal, and decimal formats. (2M)
- c) Discuss the operation of an SR flip-flop with a circuit diagram and truth table. (2M)
- d) Compare multiprocessor with multicomputer. (2M)
- e) Give the features of third generation computers. (2M)
- f) Explain the process of subtraction two signed binary numbers using 2's complement representation with example (2M)
- g) Describe the different methods of memory access (2M)
- h) How to handle internal fragmentation? (2M)
- i) List some of the most widely used secondary storage devices with their advantages. (2M)
- j) What is an Interrupt Service Routine (ISR), and what is its role in handling interrupts? (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Minimize the following Boolean function using a 3-variable K-map: (10M)
 $F(A,B,C) = \Sigma(0,1,2,3,6, 7)$ and $F(A,B,C,D) = (3,7,11,13,14,15)$
 OR
- b) What is a decoder? Draw the logic diagram for a 3-to-8 line decoder and explain its operation. (10M)

UNIT-II

3. a) What is a binary counter? Design a 4-bit binary up-counter using JK flip-flops and explain its operation. (10M)
 OR
- b) Draw and explain the block diagram of the functional units of a computer. Discuss the roles of the Input Unit, Output Unit, Central Processing Unit (CPU), Memory Unit, and Storage Unit. (10M)

UNIT-III

4. a) Explain the division of unsigned binary numbers using the long division method. (10M)
Provide an example to divide 110101_2 by 101_2 .
OR
b) What is a hardwired control unit? Explain its design and operation. Compare it with a micro programmed control unit in terms of flexibility and speed. (10M)

UNIT-IV

5. a) Describe the differences between static RAM (SRAM) and dynamic RAM (DRAM). (10M)
Explain their working principles and applications in computer systems.
OR
b) Define cache hit and cache miss. How do cache hit ratios affect the overall performance of the memory system? Provide formulas used to measure cache performance. (10M)

UNIT-V

6. a) Briefly explain about the components of hardware for connecting I/O devices. (16M)
Describe the role of interrupt hardware in it.
OR
b) Elaborate on the features of PCI bus with a neat schematic diagram. Compare it with USB. (10M)

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R23

SET - 4

II B. Tech I Semester Regular Examinations, November- 2024
DIGITAL LOGIC & COMPUTER ORGANIZATION
(Computer Science Engineering)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
3. Answer any **FIVE** Questions, each Question from each unit
from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Compare ASCII code with EBCDIC code. (2M)
- b) Convert the decimal number -45 to signed 2's complement binary representation. (2M)
- c) Distinguish between a parallel and a serial register with examples. (2M)
- d) Define system software and application software. (2M)
- e) Give the components of Von-Neumann architecture. (2M)
- f) Explain the process of subtracting two signed binary numbers using 2's complement representation. (2M)
- g) What is demand paging? (2M)
- h) Differentiate Ram with ROM. (2M)
- i) What is the purpose of secondary storage devices? Give examples. (2M)
- j) Define hit ratio. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) What is Excess-3 code? Convert the decimal number 249 to Excess-3 code. (10M)
- OR
- b) Design a full-adder circuit using basic logic gates and explain its operation with a truth table. (10M)

UNIT-II

3. a) What is the most common use of Flip-Flops? Explain about RS Flip-Flop, D-Flip-Flop and T Flip-Flop with neat diagrams. (10M)
- OR
- b) Explain the **Von-Neumann Architecture** with the help of a diagram. Discuss its major characteristics and how it differs from the **Harvard Architecture**. (10M)

UNIT-III

4. a) Use Booth algorithm to multiply 23 by 29, where each number is represented using 6 bits. (10M)

Or

- b) Describe the working of a microprogrammed control unit. How is control information stored and fetched in a microprogrammed control system? (10M)

UNIT-IV

5. a) What is cache memory? Explain its importance in a computer system and the principle of locality of reference in cache design. (10M)

OR

- b) Describe how paging works in a virtual memory system. What are page tables, and how are they used in memory address translation? (10M)

UNIT-V

6. a) Explain the different methods of accessing I/O devices in a computer system. Discuss the advantages and disadvantages of memory-mapped I/O and I/O-mapped I/O. (10M)

OR

- b) Describe the architecture of a DMA controller. Explain the steps involved in a typical DMA operation and how it transfers data between memory and I/O devices. (10M)

II B. Tech I Semester Regular Examinations, November- 2024
OOP THROUGH JAVA

(Common to
 CSE,IT,CSE(AI&ML),CSE(AI),CSE(DS),CSE(AI&DS),CSE(CS),CSE(IOT&CSIBCT),
 CSE(CS&BS),CS&BS,AI&DS,CS,AI&ML,CS&D,COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
 2. Answer **ALL** the question in **Part-A**
 3. Answer any **FIVE** Questions, each Question from each unit
 from **Part-B**
 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) List out any four popular features of JAVA that led to its success. (2M)
- b) What is meant by Data Abstraction? (2M)
- c) When a Constructor is invoked? (2M)
- d) Write any four uses of 'this' keyword. (2M)
- e) Differentiate between Arrays and Vectors in JAVA. (2M)
- f) What are the uses of Interface in JAVA? (2M)
- g) Write about JAVAs 'Object' class. (2M)
- h) Which JAVA package contains date-time API? (2M)
- i) What are the advantages of Multithreading? (2M)
- j) What is the difference between String and String Buffer class? (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Why we need OOPs in Programming language? Explain the basic principles of OOP with real time examples. (10M)
 OR
- b) When to use a Static variable in JAVA programming? Explain the importance of Static Variable with a JAVA program. (10M)

UNIT-II

3. a) How to perform Method Overloading in JAVA? Explain with a program. (10M)
 OR

- b) What are Classes and Objects in Object Oriented Programming? How are Objects created from Class? Explain with a JAVA program. (10M)

UNIT-III

4. a) Write a JAVA program to find the common elements between two arrays. (10M)

OR

- b) How JAVA implements Multiple Inheritance? Explain with suitable JAVA program. (10M)

UNIT-IV

5. a) Explain how packages are used to resolve naming conflicts in JAVA? With an example show how to add classes to packages and how to import and access the contents of packages? (10M)

OR

- b) Discuss the different ways of Reading a text file in JAVA. (10M)

UNIT-V

6. a) With a neat sketch, explain the lifecycle of a Thread in JAVA programming. (10M)

OR

- b) Explain the steps to set up JDBC development environment. (10M)

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SET - 2

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CSE(CS&BS),CS&BS,AI&DS,CS,AI&ML,CS&D,COS)

Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
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from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) What is meant by Data Encapsulation? (2M)
- b) Differentiate between 'break' and 'continue' statements. (2M)
- c) What is meant by Method Overriding? (2M)
- d) Write about 'final' keyword. (2M)
- e) What are the advantages and types of Inheritance in JAVA? (2M)
- f) Specify any three uses of 'super' keyword. (2M)
- g) List any four popular methods of JAVA's Object class. (2M)
- h) What is the purpose of the 'finally' block? (2M)
- i) What are the various states of Threads in JAVA (2M)
- j) List the life cycle methods of Java FX Application. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) How to control access to members of a class in JAVA? Explain with an example program. (10M)
- OR
- b) Write a JAVA program to illustrate the importance of precedence and Associativity of operators. (10M)

UNIT-II

3. a) Explain the special features of Constructors with a JAVA program. (10M)
- OR
- b) Write a JAVA program to demonstrate the concept of Method Overloading. (10M)

UNIT-III

4. a) What is the purpose of Interface in JAVA? Give the syntax to implement Interfaces? Explain with a JAVA program. (10M)
- OR
- b) Demonstrate the concepts of Dynamic method dispatch and upcasting in JAVA with necessary examples. (10M)

UNIT-IV

5. a) Explain the following (10M)
- i) Formatter Class
 - ii) Random class
 - iii) Temporal Adjusters Class
- OR
- b) Write a JAVA program to handle 'Arithmetic Exception' and 'Array Index Out Of Bounds Exception'. (10M)

UNIT-V

6. a) Write a JAVA program to test if a given string contains the specified sequence of char values. (10M)
- OR
- b) Explain the different ways to create Threads in JAVA with appropriate code snippets. (10M)

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R23

SET - 3

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Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
2. Answer **ALL** the question in **Part-A**
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from **Part-B**
4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) Why JAVA is Robust programming language? (2M)
- b) Write about 'private' and 'protected' access modifiers in JAVA. (2M)
- c) What is a Constructor? (2M)
- d) List any two advantages of Method Overloading. (2M)
- e) Why we need Upcasting and Downcasting in JAVA? (2M)
- f) What are abstract classes and abstract methods? (2M)
- g) Can we write finally block without the catch block in JAVA? (2M)
- h) What is a wrapper class in JAVA? (2M)
- i) How to set Thread Priorities in JAVA? (2M)
- j) Brief the steps to connect JAVA Application with Database. (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Write a JAVA program to generate the first 'n' numbers of Fibonacci series using Loop control structures. (10M)

OR

- b) Explain about various loop control statements in JAVA programming. (10M)

UNIT-II

3. a) How are object references passed to functions? Explain with a JAVA program. (10M)

OR

- b) What are objects in object oriented programming? How they are created from class? Give the syntax. (10M)

UNIT-III

4. a) Explain the representation of 2D array in computer memory. And develop a JAVA program to generate Upper and Lower triangular matrices for a given matrix. (10M)

OR

- b) Explain the steps to invoke super class constructor and methods from subclass. (10M)

UNIT-IV

5. a) Write a JAVA program to handle 'Null Pointer Exception'. (10M)

OR

- b) Write a JAVA program to read and write data to 'sample.txt' file using Random Java Access File class. (10M)

UNIT-V

6. a) Illustrate with an example, the JAVA's language-level support for synchronizing threads. (10M)

OR

- b) Explain the key components of Java FX Application Structure. (10M)

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Time: 3 hours

Max. Marks: 70

- Note: 1. Question Paper consists of two parts (**Part-A** and **Part-B**)
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 4. All Questions carry **Equal Marks**

PART -A (10x 2Marks=20M)

1. a) What is meant by Inheritance? (2M)
- b) Write the importance of 'static' keyword in JAVA. (2M)
- c) What do you mean by call by reference? (2M)
- d) Write any two special characteristics of Constructors. (2M)
- e) How do you check if an object is an instance of a particular class? (2M)
- f) Write the application of 'super' keyword. (2M)
- g) Write about JAVA formatter class and format specifiers. (2M)
- h) Write about java.lang package. (2M)
- i) List the methods used for Inter-thread communication in JAVA. (2M)
- j) What is scene graph in JavaFX? (2M)

PART -B (5x10Marks=50M)

UNIT-I

2. a) Discuss the different types of Loop structures in JAVA and write a JAVA program to display all prime numbers between 1 and 100 using any one loop construct. (10M)

OR

- b) What is the main purpose of Command line arguments in JAVA? Develop a JAVA program to demonstrate the usage of command line arguments. (10M)

UNIT-II

3. a) How to control access to members of a class in JAVA? Explain with an example program. (10M)
- OR
- b) Discuss the different ways of passing parameters to functions in JAVA. (10M)

UNIT-III

4. a) Differentiate between Abstract classes and Interfaces in JAVA. (10M)

OR

- b) Explain different types of Polymorphism supported by JAVA. (10M)

UNIT-IV

5. a) What is the significance of the CLASSPATH environment variable in creating or using a package? And explain how to set the CLASSPATH. (10M)

OR

- b) Write a JAVA program to Read and Write Binary Files using Buffered Input Stream and Buffered Output Stream. (10M)

UNIT-V

6. a) List out and explain the methods used for Inter-thread communication in JAVA. (10M)

OR

- b) With a suitable Java program, explain the usage of Class String Buffer. (10M)